16 From Flight Planning to DEM Visualisation: Improving UAV LiDAR Data for archaeological survey in Mountain Environments

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Mountain landscapes show specific challenges to UAV LiDAR survey: steep relief, dense forest canopy and limited access affect every stage of the work, from data acquisition to processing and analysis. This paper focuses on the key steps of a master's thesis workflow for generating and visualizing Digital Elevation Models (DEMs) from UAV LiDAR data. The research was developed on two case studies: Monte della Croce (Emilia-Romagna) and San Pietro in Tuba (Veneto). At both sites, several test flights were carried out with different settings to compare their outputs and identify the most suitable configuration. The point clouds were classified both in DJI Terra and in CloudCompare, so that the two classifications could be compared and an optimal point cloud obtained for further processing. The results show that flight planning has a direct impact on data quality: flight type, speed and altitude, as well as the definition of a core area over the zone of interest, determine the density and distribution of ground points and, consequently, the reliability of the final models. DEMs were then generated with WhiteboxTools at different resolutions and analyzed in QGIS through the plug-in Relief Visualization Toolbox (RVT). The readability of archaeological features and micro-topographic anomalies changed considerably depending on both the resolution of the model and the visualization applied, showing that the two parameters should be assessed together. Overall, these steps proved particularly useful for generating and interpreting UAV LiDAR data in mountain environments. By presenting them together, this paper aims to provide practical guidance for assessing the quality of these datasets and improving their interpretation in archaeological survey.